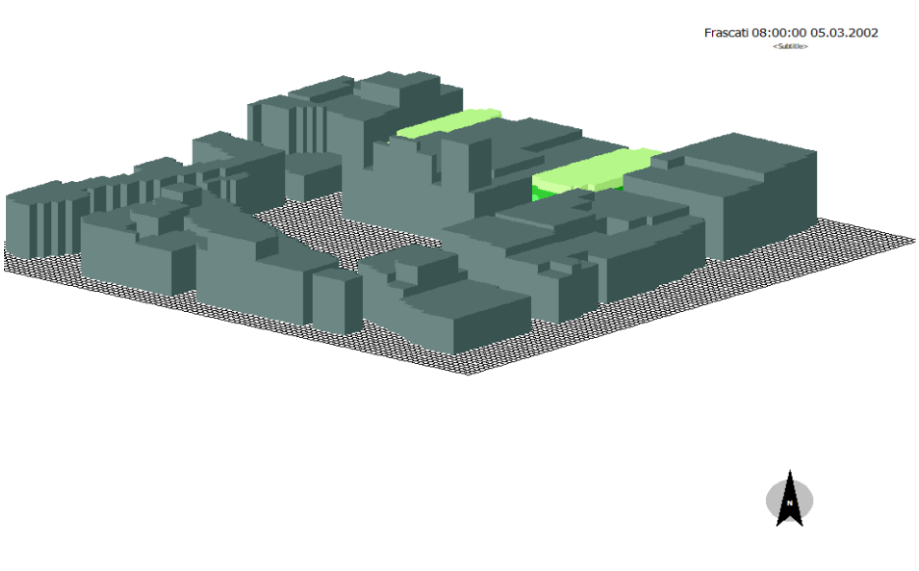
Urban Climate

ENVI Met Analysis Final Report

Department of Civil and Industrial Engineering
Course of Environmental Engineering
"Climate Change Adaptation and Mitigation"



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1.Objectives

For this simulation, the ENVI-met software was employed to generate a series of two-dimensional maps, aimed at analysing the urban environment of a selected area. The main goal was to investigate how certain variables behave in relation to the characteristics of the study area, including land cover, vegetation presence and height, building size and height, wind direction, as well as the location and intensity of pollution sources.

The variables chosen for the analysis were: Wind Speed [m/s], the rate at which air moves horizontally, Potential Temperature θ [K], defined as the temperature a particle would reach if brought adiabatically to a reference pressure p_0, in relation to the Specific Humidity [g/kg], representing the amount of water vapour per kilogram of dry air; Turbulent Kinetic Energy (TKE) [J/kg], which is mainly generated during daytime conditions within the convective layer of the Atmospheric Boundary Layer (ABL) due to buoyancy effects, leading to the formation of rising warm air parcels known as thermals; Particulate Matter Concentration PM10 [$\mu g/m^3$], used to evaluate pollution levels associated with traffic and the increase in emissions throughout the day, contributing to atmospheric turbidity (reduction of short-wave solar radiation).

The simulations were carried out for four different times over a 24-hour period, from 05 March to 06 March 2002:

1. 08:00, approximately an hour after sunrise (which occurs around 07:00 in March), corresponding to the transition from the nocturnal stable layer to the developing convective layer of the ABL.
2. Noon, when the convective layer reaches its maximum development (unstable stratification).
3. 18:00, approximately half an hour after sunset (which occurs around 17:30 in March), where we observe the development of the Stable Boundary Layer (SBL), a nighttime atmospheric layer characterized by weak turbulence and limited air mixing.
4. Midnight, where atmospheric stability is favored.

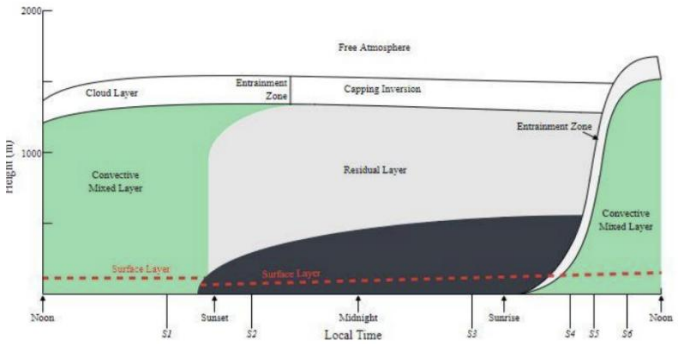


Figura 1: Atmosphere daily vertical structure

2.Study Area

The study area is located in the historic center of Frascati (RM), specifically in Piazza San Pietro, within the geographical area of the Castelli Romani. The urban area, about 22,48 km^2 lies at approximately 320 m above sea level and is set within a hilly morphological context typical of the Alban Hills.



Figura 2: Piazza San Pietro, Frascati (RM)

The historic center is characterized by a compact urban morphology, with a street network mainly composed of narrow streets, closely spaced buildings, and non-uniform building heights, which are typical features of historic urban fabrics. This configuration has a significant impact on local microclimatic conditions, influencing ventilation patterns, the distribution of solar radiation and the mechanisms of thermal exchange between built surfaces and the surrounding atmosphere. In particular, the limited width of the streets and the variability in building heights lead to prolonged shading effects and the channeling of airflow, with notable consequences for the thermo-environmental comfort of urban spaces.

From a climatic perspective, the area of Frascati falls within the temperate Mediterranean climate zone, characterized by generally mild winters and hot, dry summers. The period under analysis, corresponding to the beginning of March, represents a transitional phase between the winter and spring seasons. During this time, a gradual increase in solar radiation and daylight duration is observed, while air temperatures remain relatively moderate.

The study area, with a planimetric extent of 150 × 150 m, was implemented in the ENVI-met software within a computational domain of 100 × 100 × 35 grid cells. This configuration was adopted to ensure that the modeled buildings maintained a minimum distance of 10 m from the boundary of the computational grid, a condition required to minimize boundary effects and to improve the reliability of the simulation results. As a consequence, some buildings that are actually present in the real area were not fully represented in the digital model and were therefore only partially included in the simulation.



Figura 3: Study area from Google Earth

The selected area was also rotated by 34° toward the West in order to ensure that the central structure, San Pietro Cathedral, has its front wall aligned as straight as possible, in order to simplify the analysis.

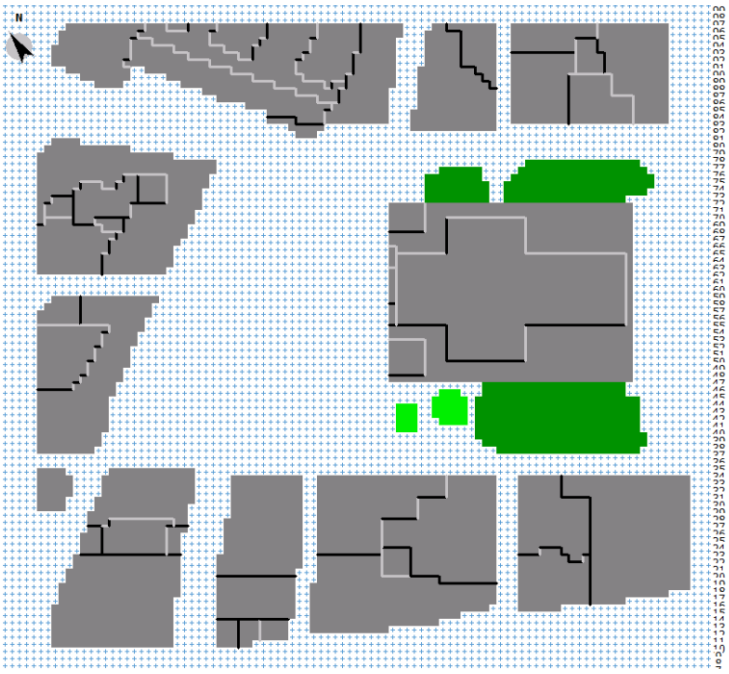


Figura 4: Digital image of the study area on ENVI-Met

The highest points of the domain are represented by the two bell towers of the Cathedral, both standing at a height of 35 meters, while the pediment reaches 32 meters. The heights of the buildings are not uniform, varying both from building to building and within the same building, due to the presence of terraces on multiple levels and internal courtyards.
The surrounding area is primarily composed of pedestrian zones paved with traditional "sanpietrini", small volcanic stone slabs such as leucitite and porphyry, which give the site its distinctive appearance. However, there are also two asphalted roads, which represent the main sources of pollution in the area.
On either side of the cathedral, tall and dense trees can be found.

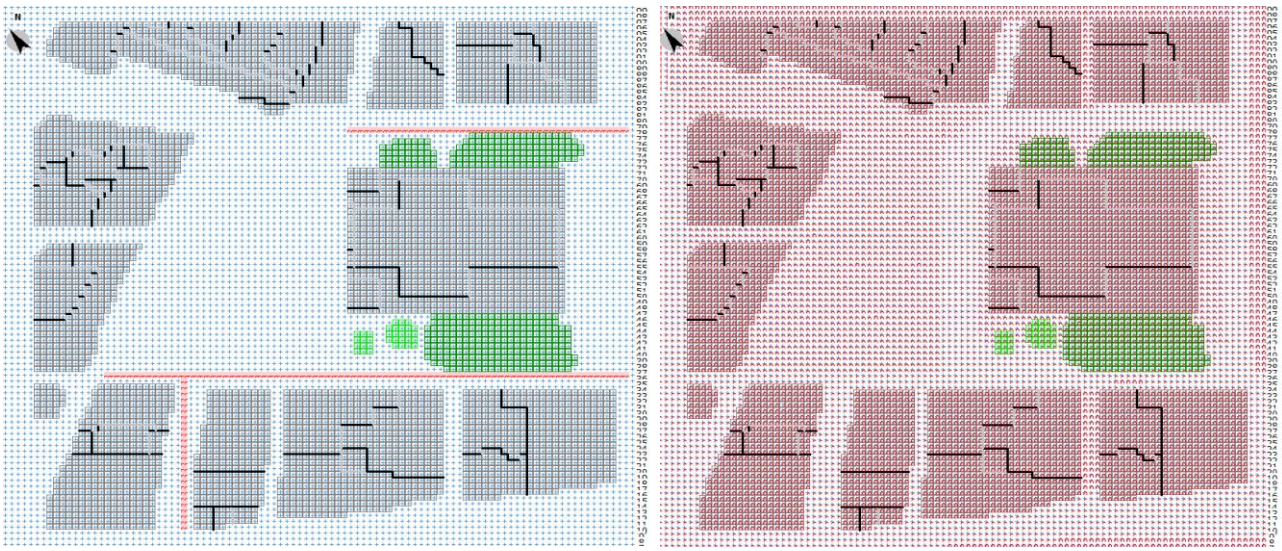


Figura 5: Pollution source and soil differentation of the study area

3.Meteorological Simulation Configuration

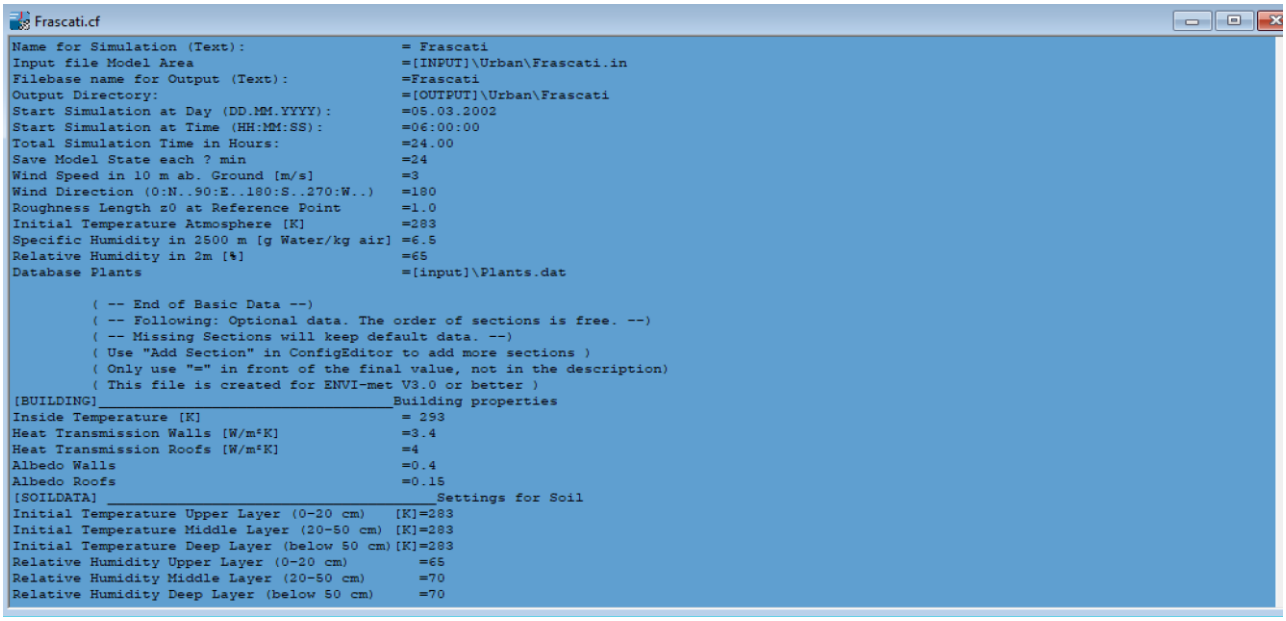


Figura 6: Study area's configuration

The simulation was initialized at 06:00 on March 5 2002, and was run for a duration of 24 hours, concluding at 06:00 on March 6. The initial atmospheric conditions for the simulation were defined coherently with the local climatic context and the urban characteristics of the study area:

- Wind Speed [m/s], evaluated at 10 meters above ground level, was set to 3 m/s. This value is representative of a compact historic center, where natural ventilation is generally limited due to high building density and urban roughness.
- Wind Direction, set at 180° (South), a methodological choice justified by the typical conditions of transitional months in the Lazio region.
- Aerodynamic Roughness Length [m], set at 1.0 m, a value indicative of a dense urban fabric, characterized by irregular surfaces that impede airflow.
- initial atmospheric Temperature [K], set at 283 K (approximately 10 °C), corresponding to typical average climate conditions in early March of the Frascati's area.
- Specific Humidity [g/kg] at 2500 m was set to 6.5 g/kg, consistent with late-winter atmospheric conditions.
- Relative Humidity at 2 m above ground was defined as 65%, a realistic value for the early hours of a late-winter day.

It should be noted that the adopted parameters do not derive from direct measurements but constitute reasonable modeling assumptions, based on the

geographical, climatic, and urban context of the study area. These assumptions enable a qualitative and comparative analysis of the microclimatic behavior of the historic center of Frascati, while maintaining physical consistency and plausibility of the results.

The configuration was further enhanced by including datas on the Heat Transmission [W/m^2K] and the Albedo of buildings, calculated considering concrete constructions. Additionally, informations on soil Temperature [K] and Relative Humidity was incorporated to account for surface-atmosphere interactions.

4.Results: analysis and comment

The analysis of meteorological variables will be conducted following an approach structured in two main phases. In the first phase, I'll examine each variable at a specific and representative time, set at 08:00, in order to assess the initial conditions and the state of the atmospheric system during the early hours of the day.
Subsequently, the analysis will be extended to the evaluation of the temporal development of each variable during the daily cycle, with the aim of describing their behavior during both daytime and nighttime hours, in order to provide an overview of the simulated meteorological variables across the full daily cycle.

4.1 Wind Velocity

The wind speed analysis was carried out with the aim of understanding the airflow behavior in a consolidated urban context charcterized by isolated buildings separated from one another by narrow alleys snd a very large central square. The analyzed output represents a horizontal section at 2 m above ground level, corresponding to the pedestrian level, at 08:00, a representative time for morning ventilation conditions and outdoor thermal comfort.
The visualization also includes flow vectors, where their verse and direction represent wind direction and their lenght represents winds speed; this allows us to interpretate the wind intensity and direction in relation to the urban morphology.

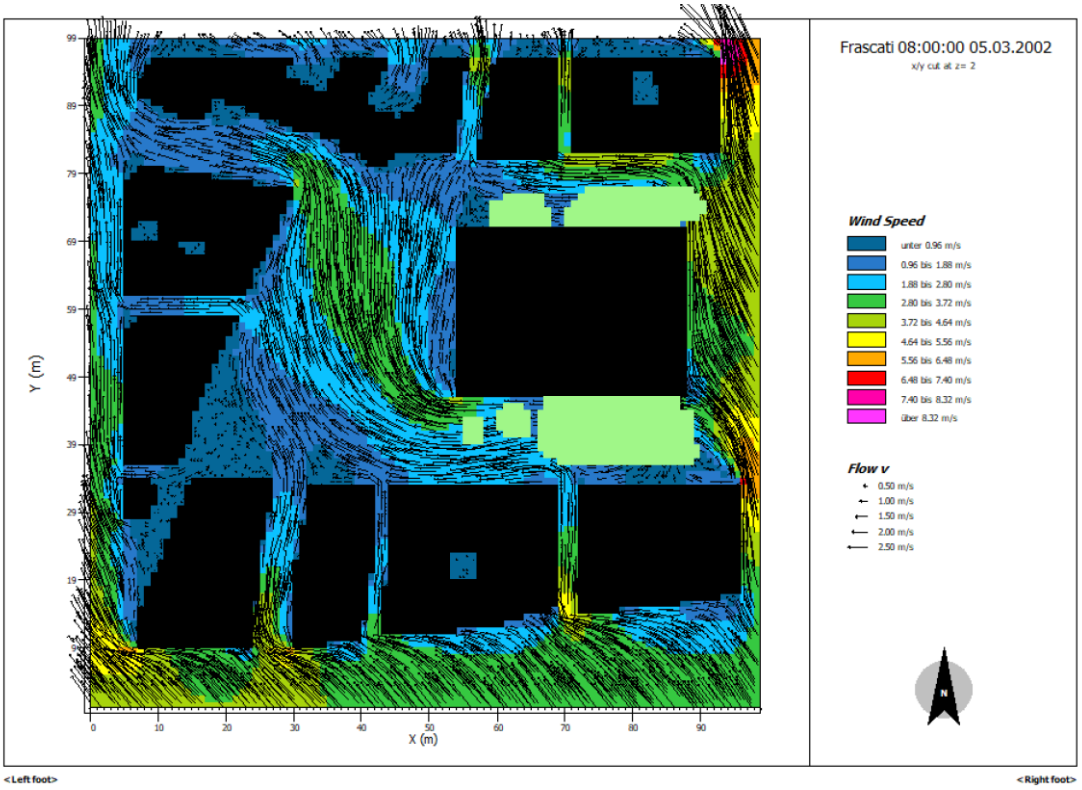


Figura 7: Wind Speed at 08.00 in x-y plan at z=2

We can observe that within the alleys and narrow passages between buildings, wind speeds are generally higher than in the surrounding areas. As the flow is forced to pass through spaces with a reduced cross-section, it becomes channelized and accelerates, producing the so-called Venturi effect. When these alleys are aligned with the prevailing wind direction, the acceleration is more pronounced.
Significantly lower wind speeds can be observed in close proximity to buildings, particularly on the leeward side of larger structures; buildings act as obstacles to the flow, blocking the incoming wind and generating wake regions characterized by flow separation and recirculation. These areas, often corresponding to poorly ventilated spaces, exhibit weak airflow.

The square exhibits a more balanced wind regime; from a microclimatic perspective, the analyzed open space acts as a ventilation lung for the surrounding urban area, functioning as a mixing zone for airflow coming from different directions. The flow lines show a progressive widening of trajectories and a smooth curvature of the wind field, indicating a gradual adjustment of the airflow to the open spatial configuration. As wind channeled through the adjacent narrow streets enters the square, an expansion of the flow occurs, comparable to an inverse Venturi effect, characterized by a sudden increase in the cross-sectional area and a consequent reduction in wind speed. Nevertheless, due to the lower degree of shielding compared to the surrounding street canyons, the open space allows for continuous air exchange, promoting moderate and relatively uniform ventilation, as highlighted by the intermediate wind speed values.

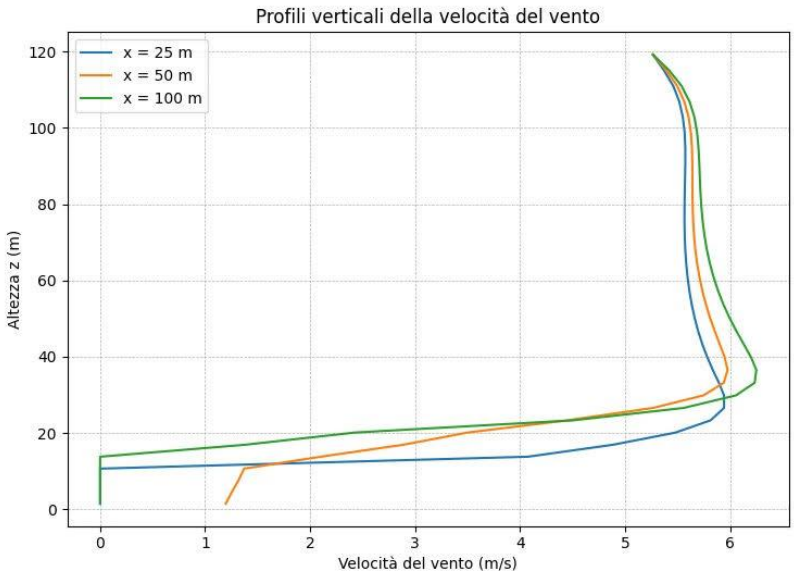


Figura 8: Vertical velocity profiles at x=25m, x=50m, x=100m

Fig. 8 represents three vertical wind speed profiles, extracted at different positions within the domain (x = 25 m, x = 50 m, and x = 100 m), that allow us to analyze the influence of urban morphology on the vertical development of wind speed.
The profile at x = 25 m shows the lowest wind speeds at pedestrian level and a rather gradual increase with height, indicating a condition strongly sheltered by buildings in the lower layers.
Then, the profile at x = 50 m, exhibits slightly higher values at low elevations and a more rapid increase in wind speed, suggesting less volumetric interference and an intermediate condition, likely associated with more open urban spaces or partial channeling of the flow.
In the end, the profile at x = 100 m displays the highest velocities throughout the vertical column representing a more exposed and less sheltered area.
Overall, the differences among the three profiles confirm that, under the same boundary meteorological conditions, the local urban configuration exerts a significant control over the vertical distribution of wind speed.

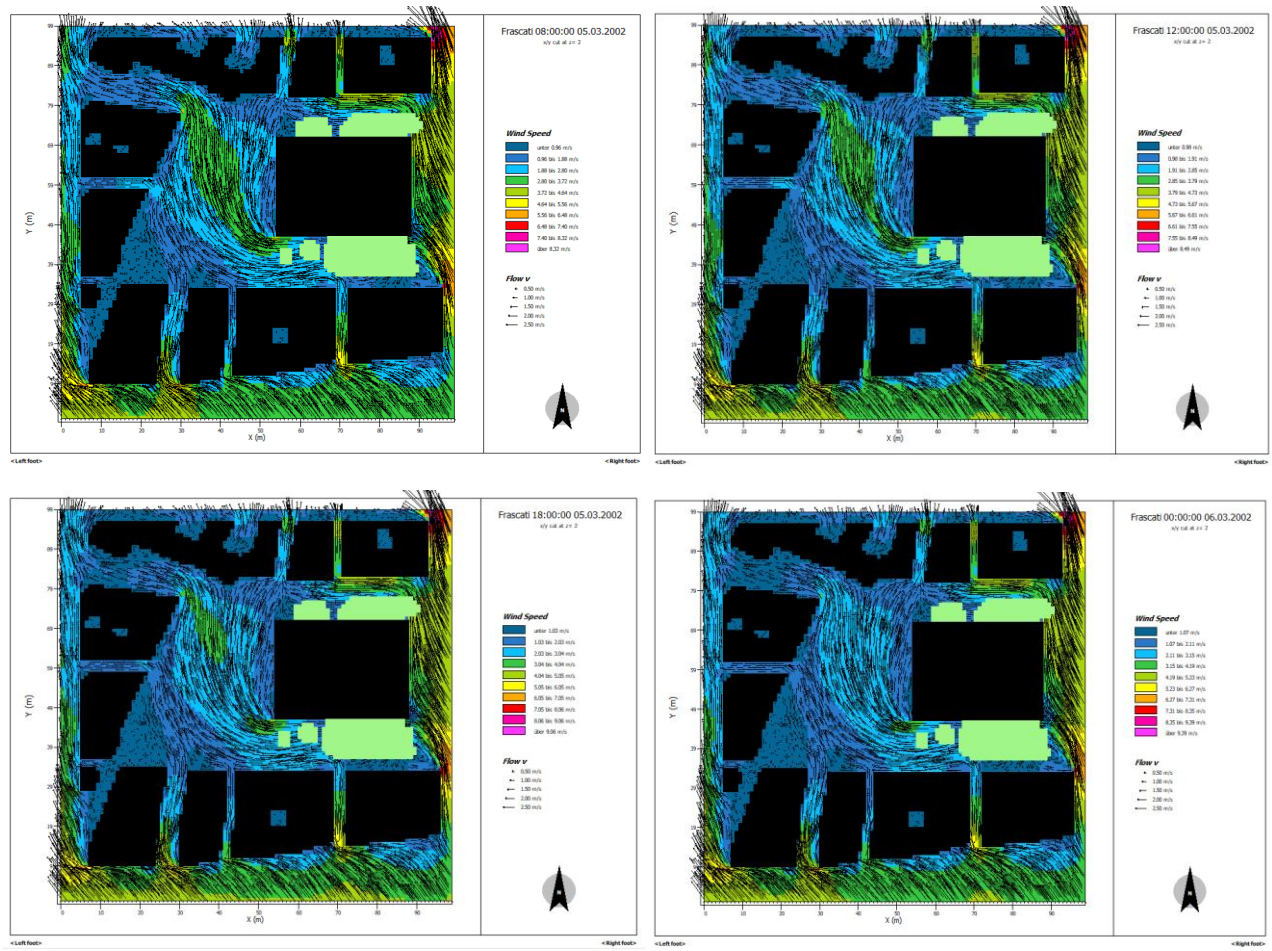


Figura 9: Wind speed daily cycle

It can be useful to analyze how the urban wind field evolves over the daily cycle:

- As we already said at 08:00, the wind field is relatively moderate and well organized. Wind speeds are generally low to intermediate, and the influence of urban geometry is clearly recognizable.
- By 12:00, the wind field becomes more energetic and spatially variable. Overall wind speeds increase, especially in areas prone to acceleration such as alleys, street intersections, and building corners. This intensification is linked to strong solar heating and the development of a convective boundary layer, which enhances vertical at pedestrian level.
- At 18:00, the wind field still reflects the daytime conditions, although signs of transition toward evening stability begin to appear. Wind speeds remain relatively high in exposed and channelized areas due to the inertia of the atmospheric boundary layer, however a gradual decrease in average wind intensity and a sharper contrast between exposed and protected areas become noticeable.
- By 00:00, the wind field weakens significantly across most of the urban area. Radiative cooling leads to the formation of a stable boundary layer; consequently, we have a dominance of low wind speeds. The square shifts toward a more weaker ventilation regime, losing much of the dynamism observed during the daytime.

From a structural point of view, the wind field remains largely consistent during the day. The same areas continue to behave in the same way: alleys as zones of acceleration and the square as a transitional space wind intermediate wind speeds. The largest variations instead occur in terms of wind intensity; across the four daily phases, the simulations show a strengthening of wind speeds from morning to midday, a persistence of relatively high speeds in the late afternoon, due to boundary layer inertia and then a marked widespread reduction during nighttime conditions.

4.2 Turbulent Kinetic Energy (TKE)

Turbulent Kinetic Energy (TKE) is a fundamental parameter for characterizing turbulence processes in urban environments, since it provides a direct measure of the intensity of chaotic fluctuations in air motion: it does not represent the magnitude of the wind velocity, but rather the degree of flow instability.
In highly anthropized contexts, the spatial distribution of TKE is mainly governed by the complex interaction between the atmospheric flow and urban morphology, reaching its maximum values in regions where the flow acceleration, strong shear and enhanced mixing occur.

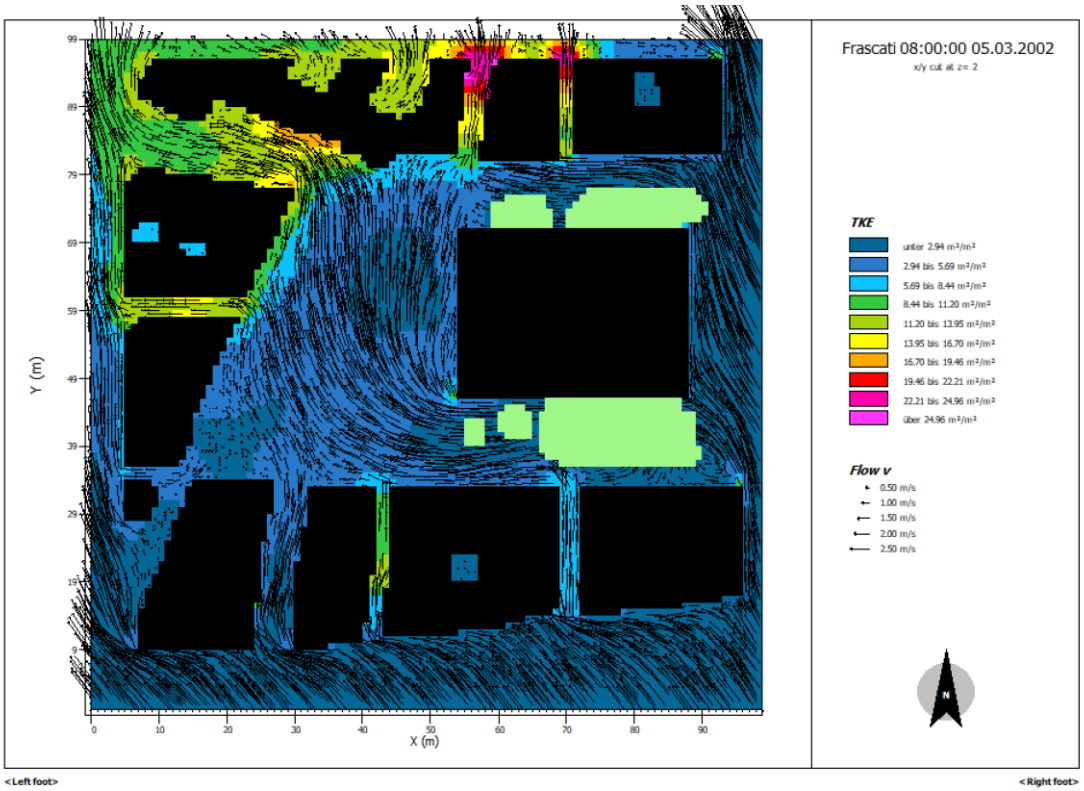


Figura 10: TKE at 08.00 in x-y plan at z=2

The analysis of the TKE map highlights that higher turbulence levels can be observed in the semi open area located in the upper left part of the domain. In this region, the incoming wind flows from a relatively open area, becomes channelled between buildings, and then suddently expands into a more open spac. For this reason the transition between open areas through a (even if moerate) local constriction, leads to flow acceleration and separation at building edges, generating vortical structures that result in elevated TKE. Locations like this represent key zones for air mixing and are characterized by intense turbulent activity.

The interior of the square is dominated by blue classes, becaused it behaves as a weakly turbulent or stagnant zone rather than as an efficient mixing zone.

In the end, we can see how high TKE values are observed only in some street canyons, particularly where the flow enters more abruptly (local maxima can be observed near the intersections of street canyons), whereas in canyons where the flow is more aligned and organized, lower turbulence levels are found.

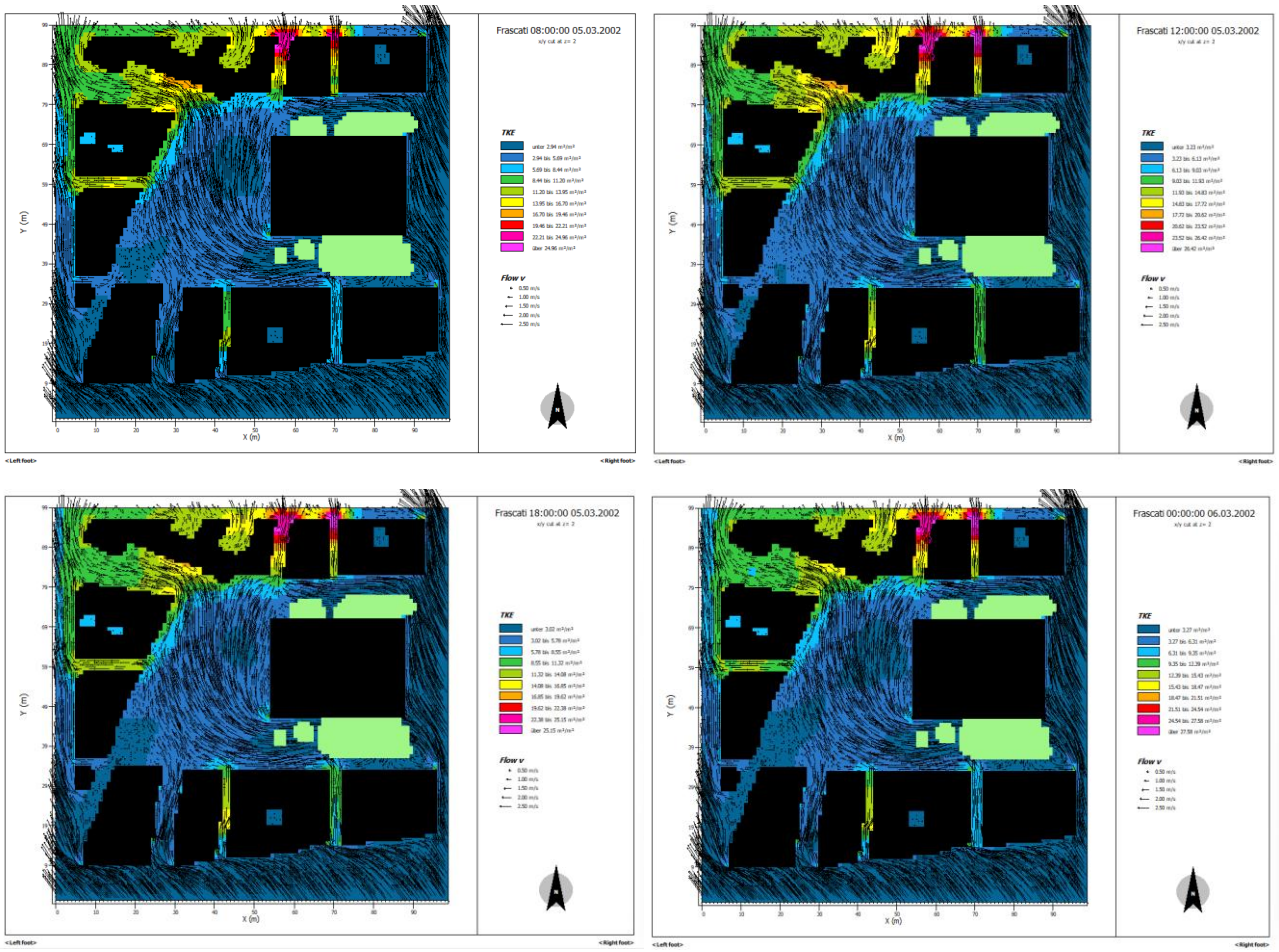


Figura 11: TKE daily cycle

TKE's evolution in the daily cycle:

- As we already said at 08:00, TKE values are generally very low across most of the domain. The flow is characterized by weak turbulence in open areas and the central square and higher values in narrow street canyons and building edges.
- By 12:00, we can observe small increases in the values throughout the area particularly along street canyons and at intersections. This increase is driven by solar heating of urban surfaces, which enhances thermal instability in the urban boundary layer and leads to turbulence production. Although the square remains a relatively low TKE region compared to surrounding canyons, its contrast with adjacent areas is reduced, indicating a more active mixing regime than in the morning.
- At 18:00, TKE remains moderate but shows a clear tendency to decrease compared to midday conditions. The spatial pattern is largely preserved, with elevated TKE still localized near confined streets and sharp geometric features, while open areas progressively return to lower values. This phase represents a transition regime in which the thermal contribution to turbulence weakens as solar forcing diminishes, and mechanical production again becomes dominant.

- By 00:00, a marked decrease in TKE values would be expected as a consequence of surface cooling, which induces a stable stratification of the atmospheric boundary layer. Under such conditions, elevated TKE levels should be confined to areas directly exposed to the incoming wind and strongly channelized flow regions. However, the simulation results show a TKE distribution that is very similar to that observed during daytime and evening hours, both in terms of spatial patterns and magnitude. This indicates that turbulence production is almost entirely governed by mechanical effects related to urban geometry, while thermal effects and the actual diurnal cycle do not emerge.

Although a diurnal cycle variation is present, the maps show only small variations in TKE because turbulence the urban domain. The contribution of thermal turbulence, that has a maximum around midday, is relatively weak at this height and mainly affects only already turbulent zones.

4.3 Potential Temperature

Potential temperature is a fundamental variable for the description of the thermal state of the air. In urban environments, its distribution is strongly influenced by building morphology, surface materials and energy exchange processes. The map has also been implementated with specific humidity contour.

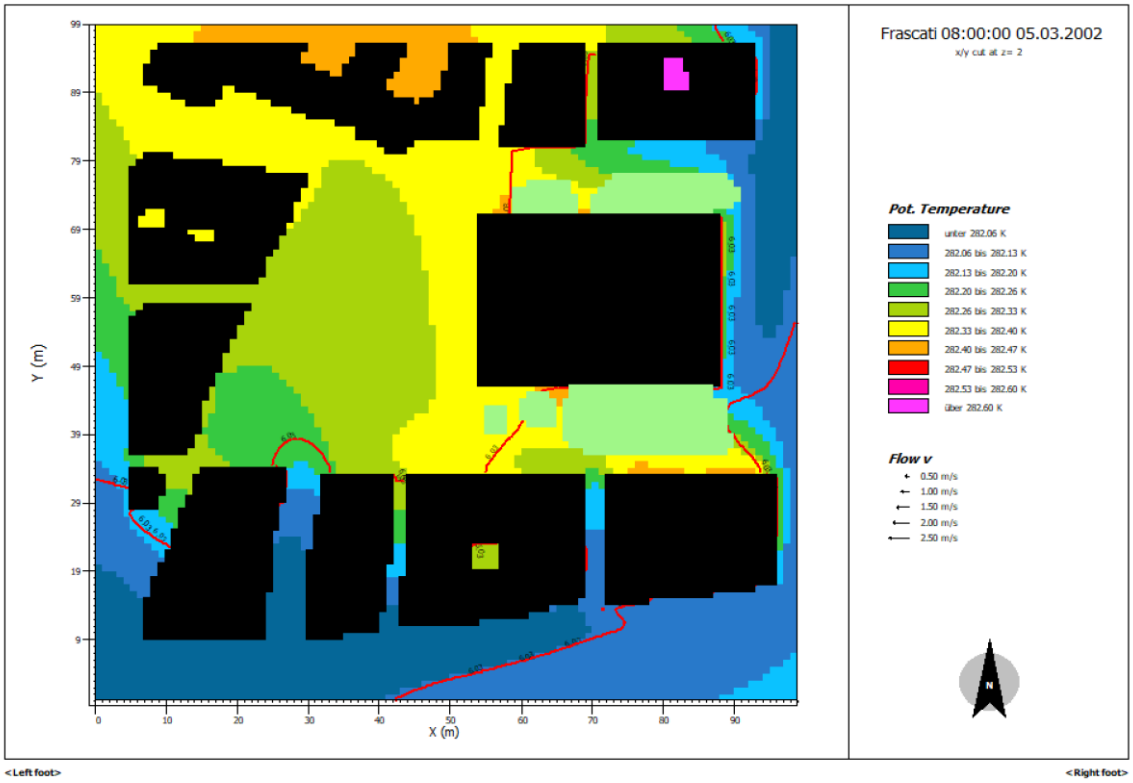


Figura 12: Potential temperature with Specific humidity contours at 08.00 in x-y plan at z=2

The horizontal section of the potential temperature field at z = 2 m is highly heterogeneous, with higher values concentrated in open and central areas of the domain, such as the square and less shielded urban corridors, and lower values occurring within street canyons and near buildings. This differentiation is mainly driven by the combined effects of shading, surface radiative balance, and urban geometry. Open areas exhibit a higher sky view factor and greater exposure to radiative fluxes, which promotes warming. Conversely, in urban canyons shading reduces solar radiative income, while geometric confinement limits exchange between air masses.

Red isolines of specific humidity represent regions of constant water vapor content and provide information on the spatial distribution of moisture independently on temperature. They tend to position themselves in transition zones between thermally distinct areas, highlighting strong differences in temperature and humidity, shaped by urban morphology and local air stagnation. This demonstrates that temperature and moisture fields are closely coupled but not spatially coincident.

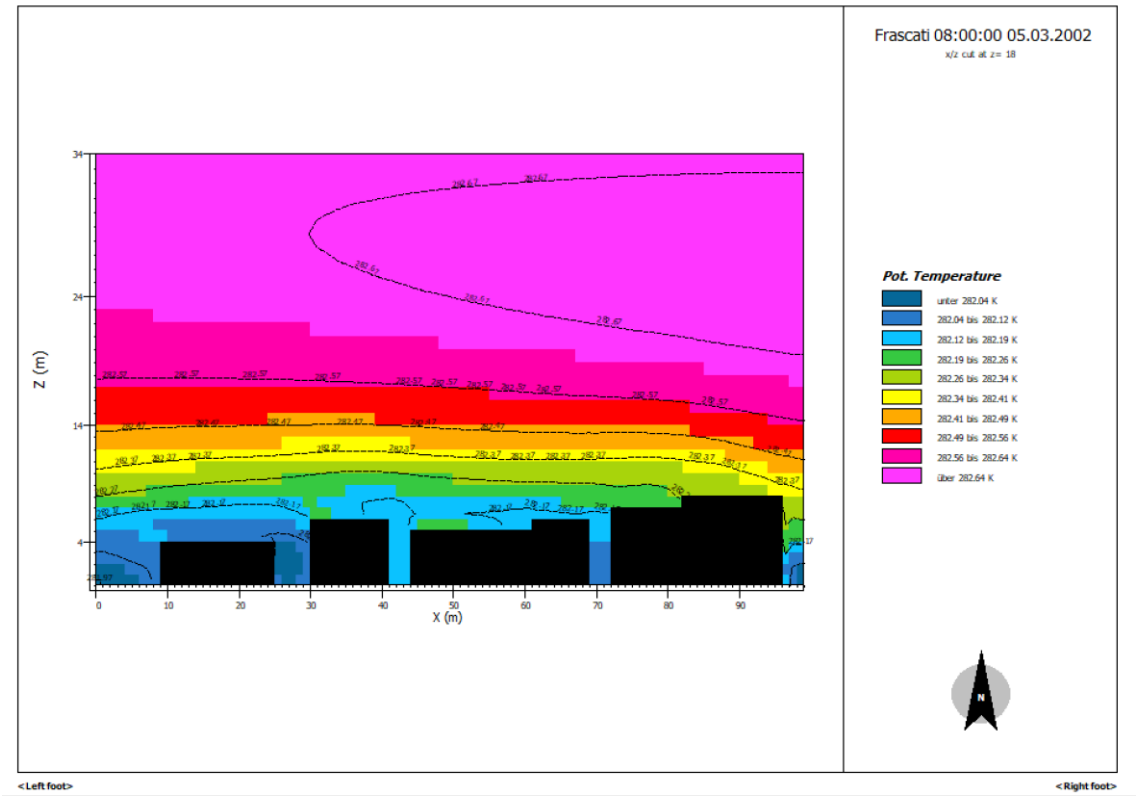


Figura 13: Potential temperature at 08.00 in x-z plan at z=18

This figure instead represents a vertical x–z section of the potential temperature field, with contour lines representing isentropic surfaces. The vertical structure clearly shows stable stratification: as expected on the theory discussed during the course, lower potential temperature values are confined near the ground, while progressively higher values occur with increasing height above the urban area. Above roof level,

isentropes are relatively horizontal, indicating a more uniform thermal structure, whereas near buildings and interstitial spaces they exhibit distortions and curvature, reflecting the direct influence of urban geometry on the lower boundary layer.

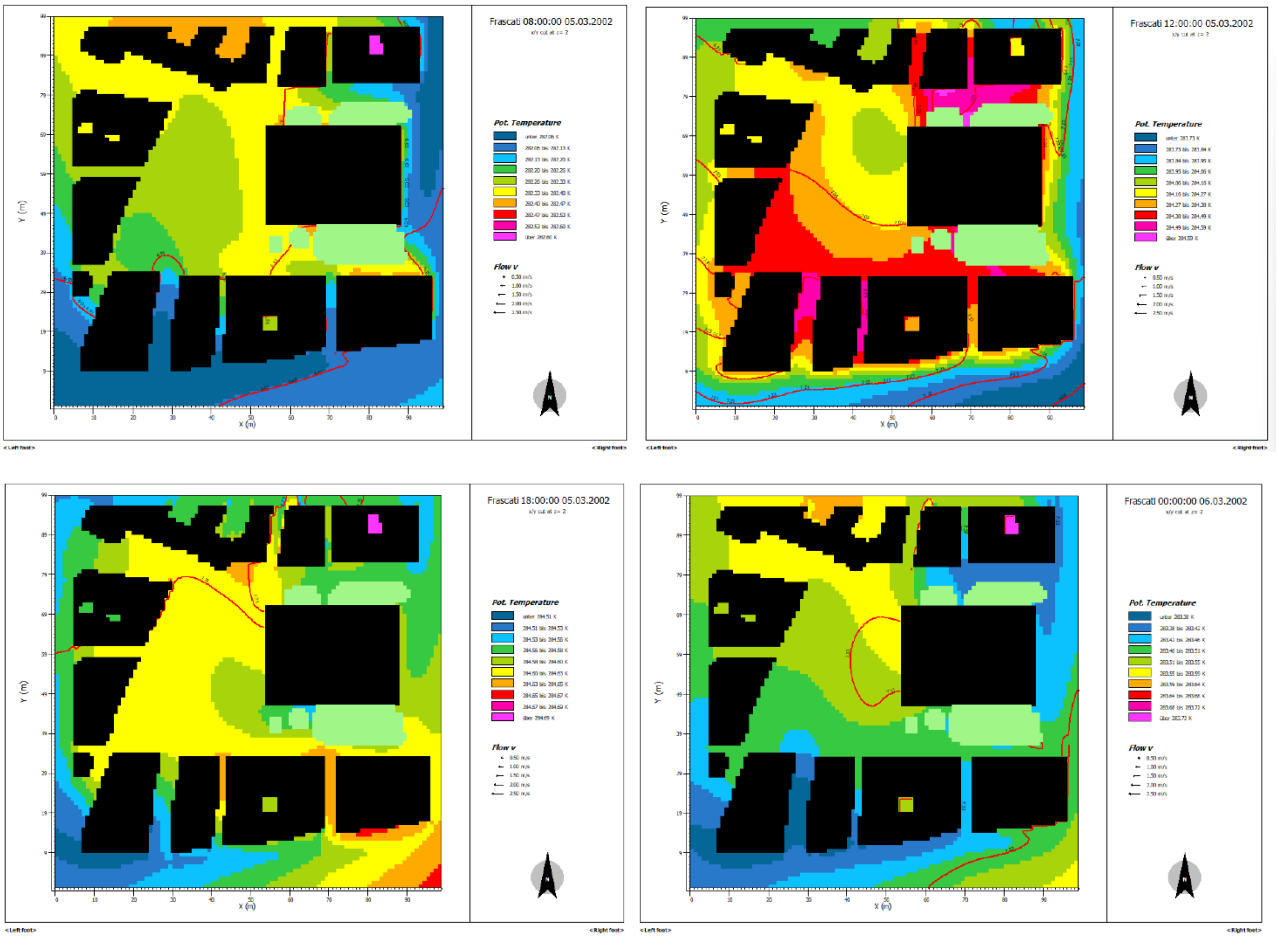


Figura 14: Potential Temperature daily cycle

Potential temperature' evolution during the daily cycle:

- As we already said at 08:00, the the potential temperature exhibits relatively low values and a strongly heterogeneous spatial distribution with cold areas are located within street canyons and along the boundaries of the domain, while more open spaces appear slightly warmer.
- Ar 12:00, a marked increase in potential temperature is observed across most of the domain. The warmest classes become dominant, especially in open areas and along more exposed urban corridors. Thermal contrasts between open spaces and street canyons intensify, producing sharp gradients near building edges. This warming phenomenon is obviously driven by solar heating. The isolines become denser and more irregular, indicating stronger specific humudity variations, due to an increase in local evaporation or in mixing and transport phenomena.
- By 18:00, potential temperature remains relatively high, but the most extreme thermal contrasts begin to weaken. Open areas still retain elevated values,

whereas gradual cooling becomes evident within street canyons. This phase represents an energetic transition, during which direct solar input decreases and the urban energy balance becomes increasingly influenced by the release of heat stored in surfaces during the day.

- At 00:00, during nighttime conditions, potential temperature decreases across the entire domain, reaching values significantly lower than those observed during daytime hours. Open areas remain slightly warmer than street canyons due to residual heat release from urban surfaces.

Temperature variations are most pronounced during daytime, when radiative heating enhances thermal contrasts between open spaces and street canyons, while nocturnal cooling leads to a more homogeneous but still structured thermal field. Specific humidity evolves coherently with temperature, showing stronger gradients during periods of intense heating and a more uniform distribution at night.

4.4 PM10 Concentration

Fine atmospheric particulate matter represents one of the most critical pollutants in urban environments, as its concentration is strongly influenced both by emission sources, such as vehicular traffic, and by atmospheric dispersion processes. In addition to its impact on air quality, PM10 is particularly harmful to human health, since these particles can be inhaled and deposited in lung tissues, causing adverse effects on the respiratory system.

From a radiative perspective, fine particulate matter has dimensions comparable to the wavelength of solar radiation, which promotes scattering phenomena. When a PM10 particle interacts with incoming solar radiation, it acts as a secondary radiation source, re-emitting the energy in all directions. Part of this radiation is reflected back into space (backscattering), another fraction contributes to the diffuse radiation reaching the Earth’s surface, while some of it may interact with other particles, triggering a chain effect of multiple scattering.

The overall result of these processes is a reduction in the amount of direct solar radiation reaching the Earth’s surface, with potential consequences for the surface energy balance and the urban microclimate.

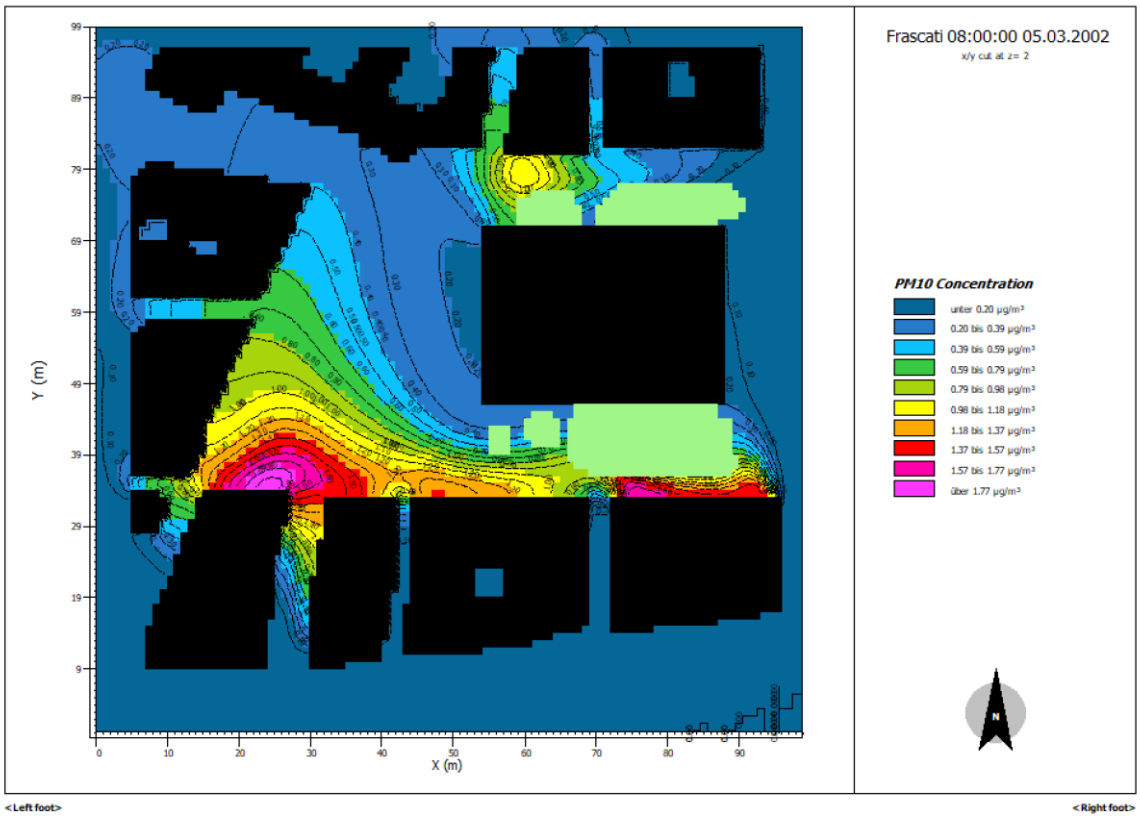


Figura 15: PM10 concentration at 08.00 in x-y plan at z=2

The distribution of PM10 concentration within the analyzed urban area exhibits strong spatial heterogeneity, closely linked to the presence of two road infrastructures and buildings' configuration. The highest concentration levels are observed along the road axes, which act ad continuous linear sources of particulate matter, mainly associated with vehicular emissions.
Local concentration maxima occur near intersections and geometric constrictions: in these areas, PM10 remains longer at pedestrian level, producing the highest concentration values shown in the map. Moving away from the main road axes, PM10 concentrations decrease rapidly, indicating more effective dilution in open and less confined spaces.

By comparing the dispersion direction of particulate matter with the prevailing wind direction, it becomes evident that the two factors are closely interconnected. In the immediate vicinity of roadways, PM10 tends to accumulate and spread preferentially along the road axis, reflecting the linear nature of traffic-related emission sources. Farther from the roads, the particulate plume progressively aligns with the wind direction, indicating that even moderate wind speeds are sufficient to redistribute particulate matter across the domain.
PM10 dispersion in the urban center is not isotropic but highly directional and controlled by urban geometry. Roads function as corridors for particulate accumulation and transport, while buildings act as barriers that limit dispersion.

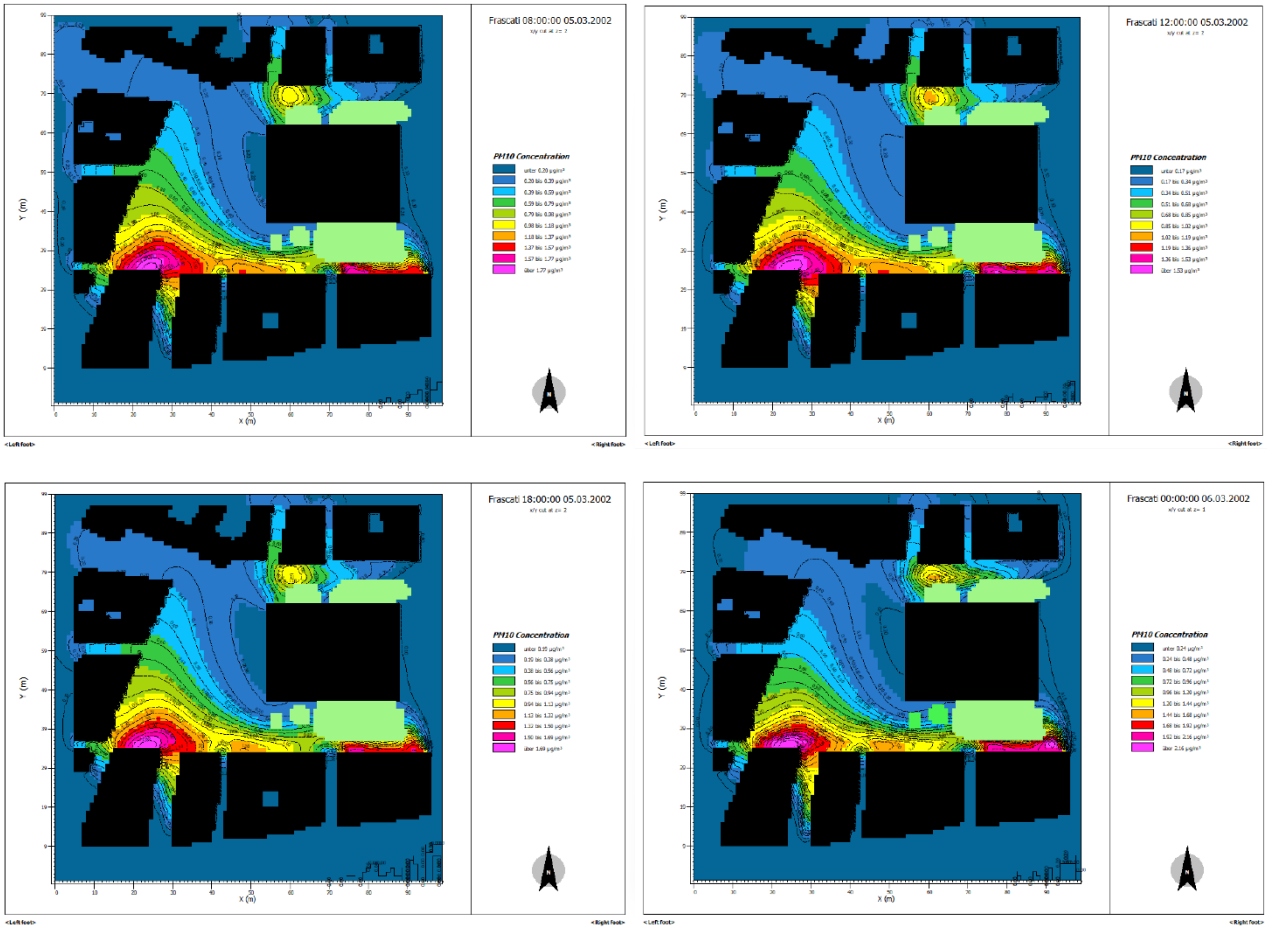


Figura 16: PM10 concentration daily cycle

PM10 concentration variations during the daily cycle:

- As we already said at 08:00, concentrations are high and strongly localized along the main road axes. The particulate plumes are well defined and elongated, with maxima near roads and intersections.
- At 12:00, a partial reduction in peak concentrations is observed, accompanied by a broader spatial extent of the plumes. PM10 appears more diluted than in the morning, although concentration maxima remain anchored to the main roads. This evolution is consistent with increased atmospheric turbulence and enhanced vertical mixing induced by solar heating, which promote PM10 dispersion and reduce local accumulation despite the continued presence of emission sources.
- By 18:00, the decrease in solar radiation leads to weakened turbulence, while evening traffic contributes to renewed emissions. As a result, the plumes become more compact and concentration gradients does not vary significantly compared to the previous hours.
- At 00:00, PM10 show defined but more confined maxima. Atmospheric conditions become more stable, vertical mixing is minimal, and particulate

matter tends to stagnate at pedestrian level. Dispersion is therefore strongly limited and controlled almost exclusively by urban geometry.

PM10 concentration varies only slightly throughout the day because it is mainly controlled by factors that are nearly constant over time. Roads, which represent the primary sources of PM10, are active for most of the daily cycle, and buildings form urban canyons that strongly limit dispersion. The wind, weak and channeled along streets, transports particulate matter without effectively diluting it, while the day–night variations in turbulence are insufficient to overcome this confinement. Consequently, PM10 tends to stagnate in the same urban sectors, exhibiting only local and moderate variations over the course of the day.

5.Conclusion

In conclusion, the simulation made it possible to analyze in detail the interactions between urban morphology and atmospheric conditions within a complex context such as Piazza San Pietro in the historic center of Frascati. The results clearly highlight the decisive role of the compact urban structure in modulating ventilation, temperature distribution, specific humidity, and turbulence levels. In particular, the presence of closely spaced buildings, narrow streets, and non-uniform building heights promotes wind channeling effects, prolonged shading areas, and reduced pollutant dispersion, especially during the evening and nighttime hours, when atmospheric stability conditions prevail.

The information obtained can provide a useful basis for future analyses aimed at sustainable urban planning and the improvement of environmental comfort in public spaces, while respecting the morphological and historical specificities of the urban context.